

SettleMatch

Complete Project Manual

Reconcile Razorpay settlements to bank credits - safely.

Part A: What the project is

Part B: Architecture and tech stack

Part C: Step-by-step UI manual with live screenshots

Part D: Your batch results and how to run

Rules first. AI second. Gate always.

A1. What is SettleMatch?

SettleMatch is an AI Finance Controller for Razorpay AI Buildathon Track 04. It solves one real finance-ops job: take a Razorpay settlement export and a bank statement, automatically link each settlement row to the correct bank credit, report how much money was recovered, and list every row that could NOT be matched safely - with reasons and an audit trail.

The business problem

- Merchants get settlements in Razorpay exports (CSV/PDF/Excel).
- The same money appears as credits in the bank statement - different format, different narration.
- Finance teams match these manually in Excel - slow, error-prone, hard to audit.
- Generic AI chatbots guess links; finance needs verification, not generation.

What SettleMatch does differently

- Rules engine handles clear cases first (UTR match, amount + date window).
- Google Gemini handles messy leftovers - only from a short candidate list.
- Confidence gate (default 85%): low-confidence AI picks become exceptions, not auto-approves.
- Collision detector blocks ambiguous double-matches.
- Human review queue only for exceptions - not for every row.

Who it is for

- Merchant finance / ops teams reconciling daily settlements.
- Razorpay internal tooling mindset - trust, audit, measured accuracy.

What you built (deliverables)

- Streamlit product UI with 7 pages: Upload, Connections, Dashboard, Matches, Exceptions, Simulator, Export.
- Python pipeline: extract, rules, AI, gate, reconcile, export.
- Multi-format ingest: PDF, Excel, Word, CSV, TXT.
- Eval harness with ground-truth precision/recall metrics.
- Audit pack export: PDF + Word + ZIP.

B1. Tech Stack

Layer	Technology
Language	Python 3
UI	Streamlit
Data	Pandas
AI	Google Gemini (gemini-3.6-flash)
Charts	Altair
PDF extract	pdfplumber
Excel	openpyxl
Word	python-docx
Reports	fpdf2
Config	python-dotenv

Production roadmap (Connections tab): Razorpay API + bank SFTP/webhook on a nightly schedule. Not in demo scope: React, PostgreSQL, Docker.

B2. Pipeline Architecture

Upload PDF/Excel/Word/CSV/TXT

- > Extract strict columns only
- > Rules: UTR exact match; amount + date window
- > Collision check on ambiguous UTR/amount
- > Gemini on unmatched leftovers (candidate list only)
- > Confidence gate ≥ 0.85
- > Output: Matches | Exceptions | Audit log
- > Export PDF / Word / audit ZIP

B3. Your Batch Results (from localhost run)

After uploading settlements.csv (100 rows) and bank.csv (125 rows) and clicking Reconcile, SettleMatch produced the following metrics:

Metric	Value
Settlements processed	100
Bank rows ingested	125
Match rate	90.0%
Matched total	90
Rules matched	75
AI matched (gated)	15
Exceptions (honest refusals)	10
Collisions blocked	15
Money matched	Rs 330,754.73
Money at risk	Rs 47,773.55
Recovery rate	87.4%
Eval precision	100%
Eval recall	100%
False matches	0

C1. Manual Walkthrough - Before You Start

Step 1

Open terminal in the RazorHack folder.

Step 2

Activate venv: `.\venv\Scripts\Activate.ps1`

Step 3

Run: `streamlit run app.py`

Step 4

Open browser: `http://localhost:8501`

Step 5

Keep sample files ready: `data/samples/settlements.csv` and `data/samples/bank.csv` (or your own PDF/Excel exports).

C2. Page 1: Upload

Step 1

Sidebar mein Upload select karo. Yeh starting point hai - yahan se reconciliation shuru hoti hai.

Step 2

Left side: do file uploaders - Settlements export (Razorpay) aur Bank statement. Koi bhi format chalega: PDF, Excel, Word, CSV, TXT.

Step 3

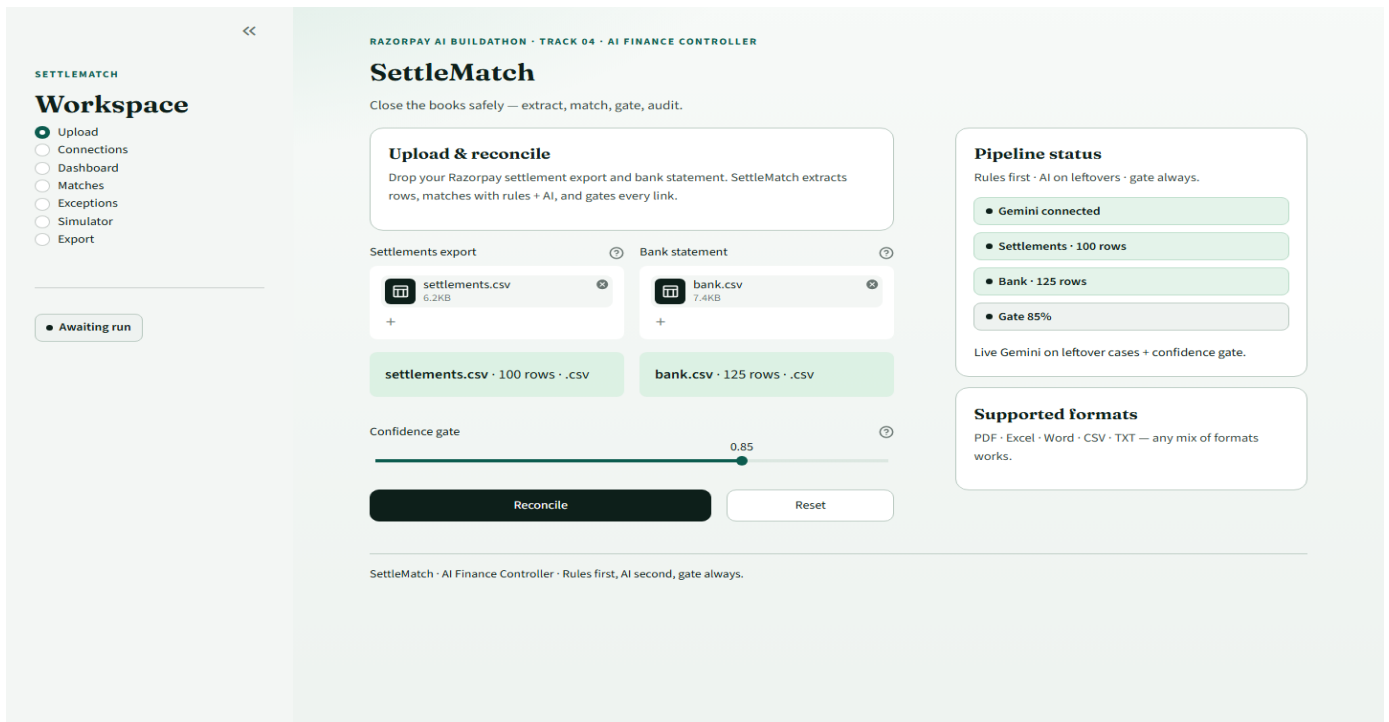
Files upload karte hi neeche green box dikhega: filename + kitni rows extract hui (example: `settlements.csv` - 100 rows - `.csv`).

Step 4

Confidence gate slider set karo (default 0.85). Isse neeche AI matches auto-approve nahi honge.

Step 5

Dono files valid hone par Reconcile button enable hoga. Click karo - pipeline chalegi (extract -> rules -> AI -> gate -> audit). Complete hone par auto Dashboard khulega.



Upload page: settlements.csv (100 rows) + bank.csv (125 rows) loaded, Reconcile ready.

C3. Page 2: Connections (Production Path)

Step 1

Sidebar se Connections kholo. Yeh dikhata hai production mein manual upload ki jagah automatic ingest kaise hoga.

Step 2

Razorpay test-mode Key ID + secret - nightly settlements API pull.

Step 3

Bank SFTP path ya Webhook URL - bank credits automatically ingest.

Step 4

Schedule: Daily 2:00 AM IST + Auto-run toggle. Last run Auto OK dikhega.

Step 5

Pitch mein bolo: Upload demo ke liye hai; real companies Connections se auto-reconcile karti hain.

The screenshot shows the SettleMatch dashboard interface. On the left is a 'Workspace' sidebar with navigation links: Upload, Connections (selected), Dashboard, Matches, Exceptions, Simulator, and Export. Below these is a status indicator 'Awaiting run'. The main content area is titled 'SettleMatch' with the tagline 'Close the books safely — extract, match, gate, audit.' and a sub-header 'RAZORPAY AI BUILDATHON · TRACK 04 · AI FINANCE CONTROLLER'. The 'Connections' section describes the production ingest process. It includes a 'Data sources' box, a 'Razorpay (test mode)' section with fields for Key ID (rzp_test_SettleMatch) and Key secret, and a 'Bank ingest' section with 'Ingest method' (SFTP selected) and 'SFTP path' (sftp.hdfcbank.com/settlematch/incoming). A 'Schedule' section shows 'Hour' (02:00) and 'Minute' (0) with an 'Auto-run' toggle. On the right, a 'Live status' box shows 'Razorpay · test mode connected', 'SFTP · sftp.hdfcbank.com/settlematc...', 'Daily 02:00 IST', and 'Auto reconcile · ON'. Below this, 'LAST RUN' shows 'Auto ✓' on '23 Aug 2026, 02:00 AM', and 'NEXT RUN' shows 'Scheduled' on '24 Aug 2026, 02:00 AM'.

Connections: Razorpay test API, bank SFTP, daily schedule, last auto-run status.

C4. Page 3: Dashboard (After Reconcile)

Step 1

Reconcile ke baad Dashboard automatically khulta hai. Yahan finance reviewer ko poora picture milta hai.

Step 2

Top KPIs: Match rate 90.0%, Recovery 87.4%, Money matched Rs 330,755, Money at risk Rs 47,774.

Step 3

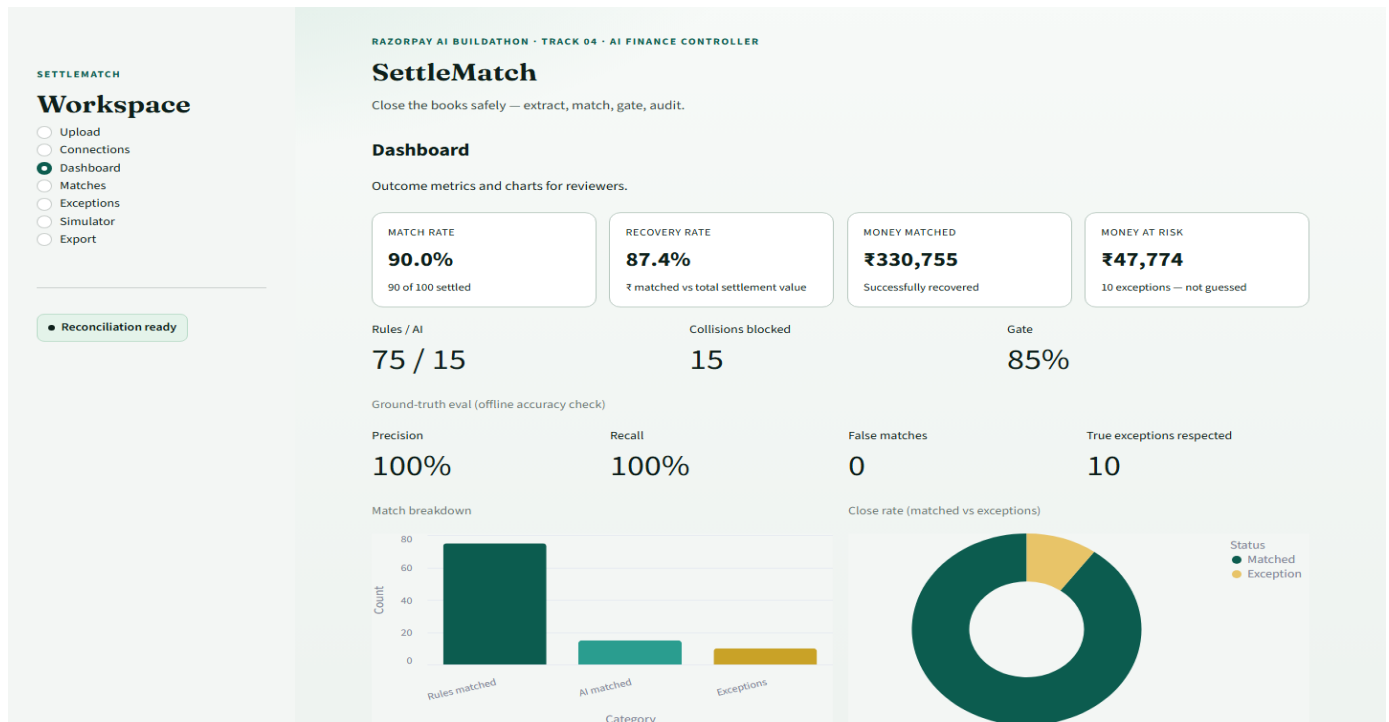
Breakdown: 75 rules + 15 AI matches. 10 exceptions honestly listed.

Step 4

Charts: match breakdown bar, donut close rate, money recovered vs at risk, rules vs AI pie.

Step 5

Eval strip: precision/recall 100%, 0 false matches - ground truth se verified.



Dashboard with live data: 90 matched, 10 exceptions.

C5. Page 4: Matches and Explainability

Step 1

Sidebar se Matches kholo. Saari accepted pairings table mein dikhti hain.

Step 2

Columns: settlement_id, bank_txn_id, method (rule / ai_gated), confidence score.

Step 3

Neeche dropdown se koi bhi settlement select karo - step-by-step explanation dikhega.

Step 4

Example: 'UTR exact match' ya 'AI picked candidate 3 at 0.91 confidence, passed gate'.

Step 5

Pitch point: har paisa link explainable hai - black box nahi.

SETTLEMATCH

Workspace

☐ Upload
 ☐ Connections
 ☐ Dashboard
 ☒ Matches
 ☐ Exceptions
 ☐ Simulator
 ☐ Export

● Reconciliation ready

RAZORPAY AI BUILDATHON · TRACK 04 · AI FINANCE CONTROLLER

SettleMatch

Close the books safely — extract, match, gate, audit.

Matches & explainability

Accepted pairings + why each decision was made.

settlement_id	bank_txn_id	method	confidence	reason
SET-001	BK-0001	rule_exact_utr	1	Exact UTR UTR3321819600 and amount match
SET-002	BK-0002	rule_exact_utr	1	Exact UTR UTR9083863794 and amount match
SET-003	BK-0003	rule_exact_utr	1	Exact UTR UTR6542351161 and amount match
SET-004	BK-0004	rule_exact_utr	1	Exact UTR UTR4078161849 and amount match
SET-005	BK-0005	rule_exact_utr	1	Exact UTR UTR0341316475 and amount match
SET-006	BK-0006	rule_exact_utr	1	Exact UTR UTR4192832764 and amount match
SET-007	BK-0007	rule_exact_utr	1	Exact UTR UTR5030564139 and amount match
SET-008	BK-0008	rule_exact_utr	1	Exact UTR UTR7672423884 and amount match

Explain a match

Settlement

SET-001

Stage: Rules engine

Method: rule_exact_utr

Confidence: 1.0

Steps

Bank txn: BK-0001

Summary: Exact UTR UTR3321819600 and amount match

Matches table + explainability panel for selected settlement.

C6. Page 5: Exceptions and Human Review

Step 1

Exceptions page par 10 refused rows dikhengi - yeh guessed nahi, honestly refused.

Step 2

Har exception ke saath reason: low confidence, collision, no candidate, etc.

Step 3

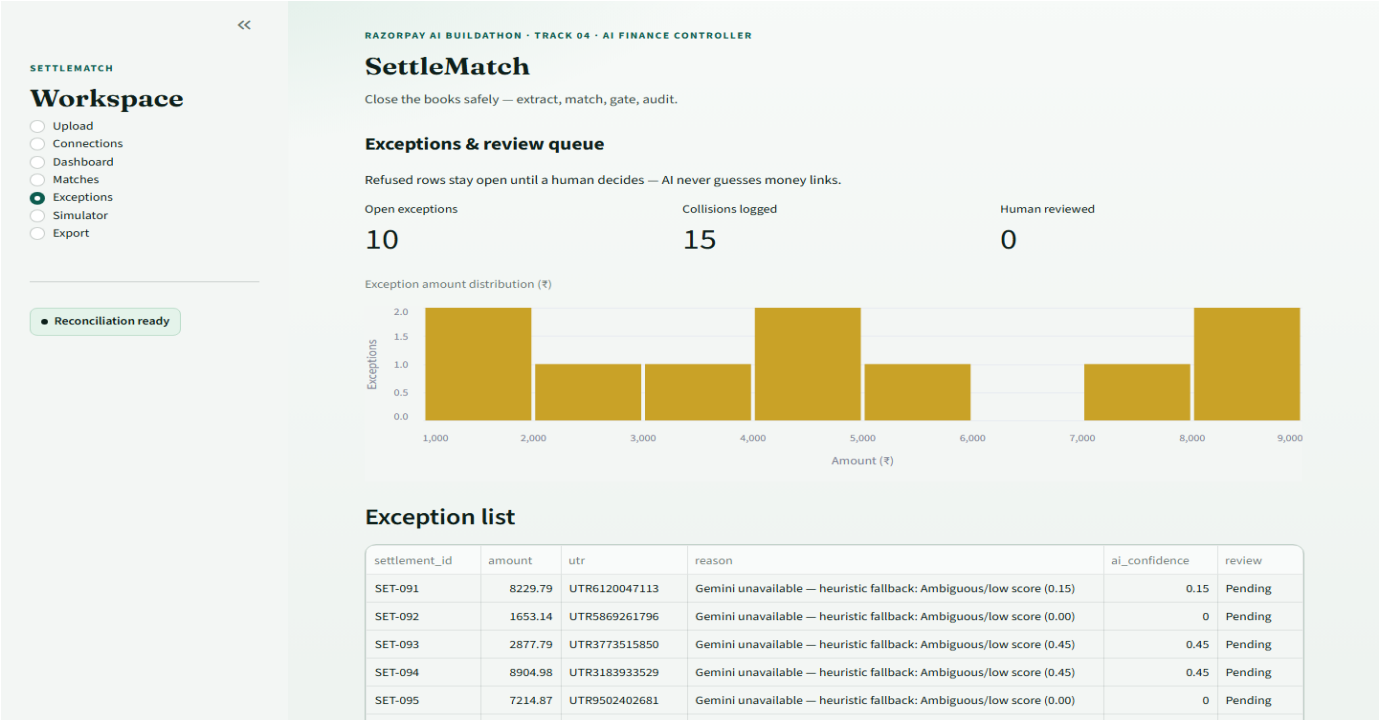
Amount distribution chart - kitna paisa at risk hai.

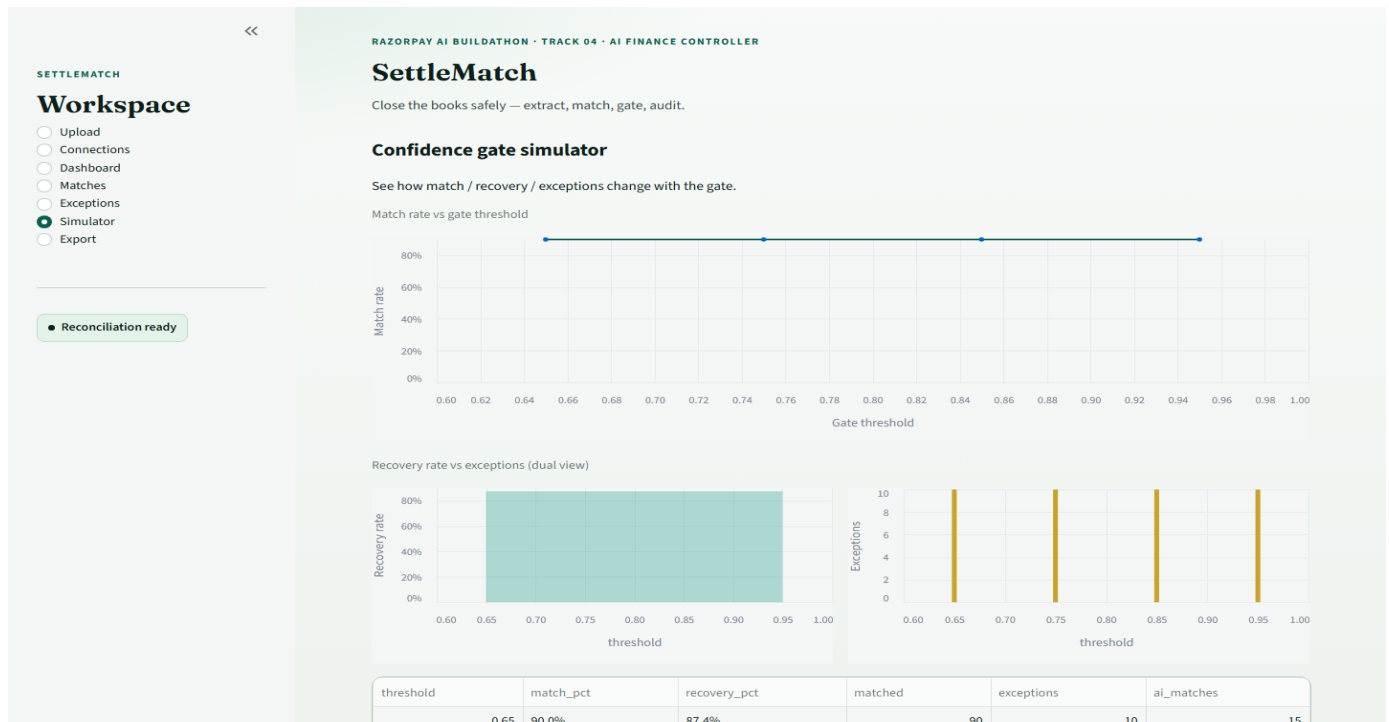
Step 4

Human review buttons: Approve manually / Reject / Needs finance follow-up.

Step 5

Finance team sirf yahan baithti hai - baaki sab automatic.





Gate simulator charts after reconciliation data loaded.

C8. Page 7: Export and Audit Pack

Step 1

Export page se reviewer ke liye downloads milte hain.

Step 2

PDF report - summary stats + matches + exceptions.

Step 3

Word report - same content, editable.

Step 4

Audit pack ZIP - matches.csv, exceptions.csv, audit_log.jsonl, report.json.

Step 5

Matches CSV - spreadsheet import ke liye.

SETTLEMATCH

Workspace

Upload

Connections

Dashboard

Matches

Exceptions

Simulator

Export

● Reconciliation ready

RAZORPAY AI BUILDATHON · TRACK 04 · AI FINANCE CONTROLLER

SettleMatch

Close the books safely — extract, match, gate, audit.

Export & audit pack

Download reports for submission and reviewers.

Sources: settlements.csv · bank.csv

▼ {

"total_settlements" : 100

"matched" : 90

"exceptions" : 10

"collisions_detected" : 15

"match_rate" : 0.9

"rule_matches" : 75

"ai_matches" : 15

"money_matched_inr" : 330754.73

"money_at_risk_inr" : 47773.55

"total_settlement_inr" : 378528.28

"recovery_rate" : 0.8738

"ai_mode" : "gemini"

"confidence_threshold" : 0.85

}

PDF report

Word report

Audit pack (ZIP)

Matches CSV

Audit trail

ts	settlement_id	bank_txn_id	stage	method	accepted	confidence	reason
2026-08-23T08:52:12.414867+00:00	SET-001	BK-0001	rules	rule_exact_utr		1	Exact UTR UTR3321819600 and amount matc

Export buttons + audit trail table with all decisions.

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D1. How to Run (Copy-Paste)

```
cd RazorHack
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
copy .env.example .env
python scripts/generate_sample_docs.py
streamlit run app.py
# Browser: http://localhost:8501
# Upload -> both files -> Reconcile -> Dashboard
```

D2. Repository Map

File / Folder	Purpose
app.py	Streamlit UI - all 7 pages
src/reconcile.py	Main pipeline orchestrator
src/extract.py	PDF/Excel/Word/CSV/TXT extraction
src/rules.py	UTR + amount/date rule matcher
src/ai_matcher.py	Gemini + heuristic fallback
src/gate.py	Confidence threshold gate
src/export_report.py	PDF/Word/ZIP export
eval/run_eval.py	Ground-truth accuracy test
data/samples/	Demo upload files
docs/screenshots/	UI screenshots for this guide

D3. 5-Minute Pitch Script

- 0:00 - Problem: do lists, manual match, audit pain.
- 0:45 - Upload live files, click Reconcile.
- 1:30 - Dashboard: 90% match, Rs 3.3L recovered, 0 false matches.
- 2:30 - One exception + human review - we refuse, never guess.
- 3:30 - Connections: production auto-ingest story.
- 4:30 - Download audit pack. Close.